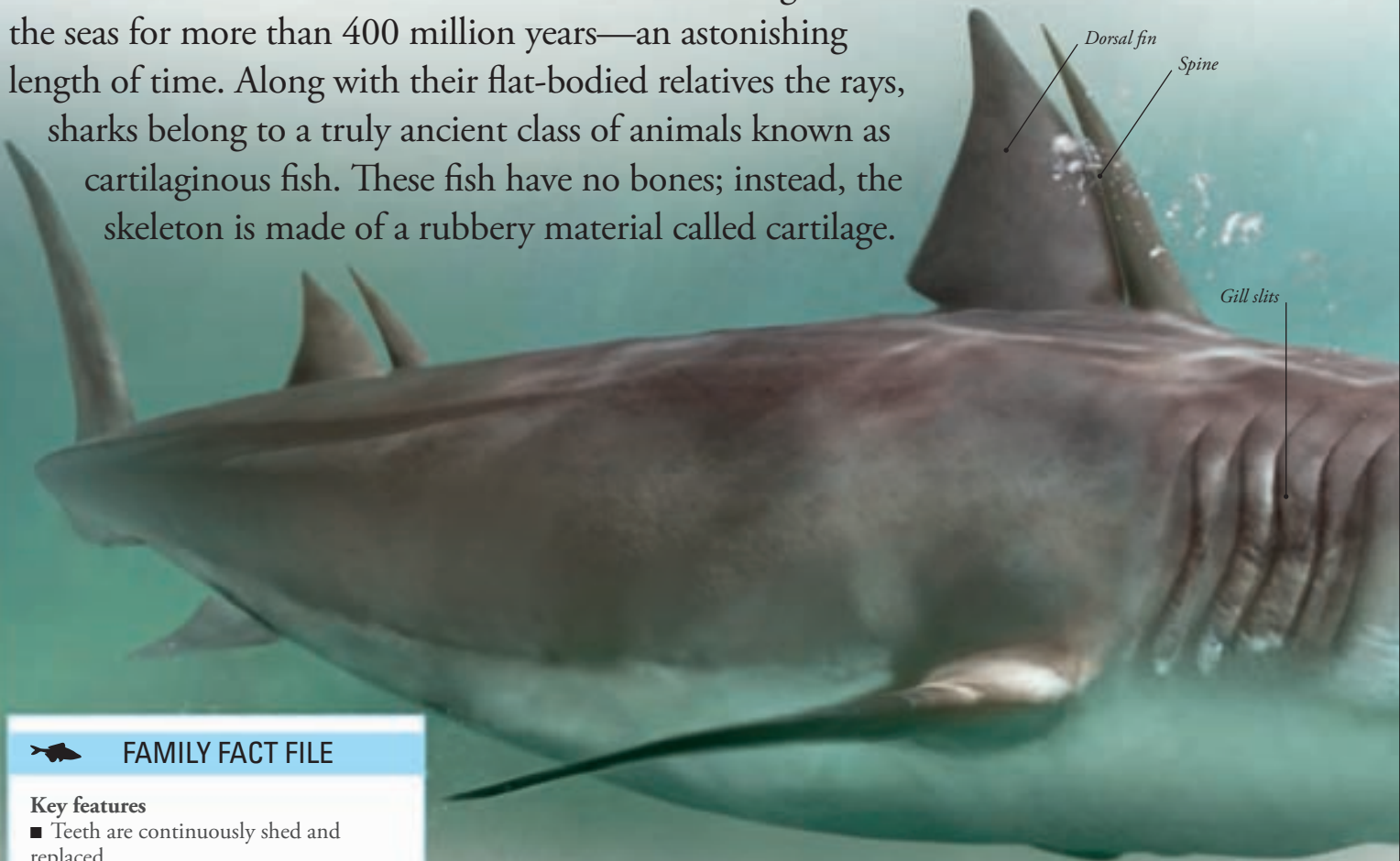


Sharks and rays

Fossil teeth reveal that killer sharks have been cruising the seas for more than 400 million years—an astonishing length of time. Along with their flat-bodied relatives the rays, sharks belong to a truly ancient class of animals known as cartilaginous fish. These fish have no bones; instead, the skeleton is made of a rubbery material called cartilage.



FAMILY FACT FILE

Key features

- Teeth are continuously shed and replaced
- Skeleton made of cartilage
- No ribs whatsoever
- No air bladder for controlling buoyancy
- Sharks must keep swimming or they will sink
- Fins in pairs for steering, unlike earlier fish

When

The earliest known fossils of sharks and rays date back to the Late Silurian, almost 420 million years ago.

Hybodus

hy-BODE-us

- **When** Late Permian to Late Cretaceous
- **Fossil location** Europe, N. America, Asia, Africa
- **Habitat** Oceans
- **Length** 6 ft (2 m)
- **Diet** Small marine animals

Hybodus looked as fierce as any modern shark and had the classic streamlined shape, but its teeth and fins were different from today's

sharks. It had two types of tooth: sharp ones at the front for seizing slippery prey such as fish, and flatter, more blunt teeth at the back of the mouth for crushing shells. In front of *Hybodus*'s dorsal fin (the fin on its back) was a long, bladelike spine. This may have helped the fin to cut through the water more easily or it may have been used for defense.

Heliobatis

he-lee-oh-BAT-iss

- **When** 54–38 million years ago (Early to Middle Paleogene)
- **Fossil location** USA
- **Habitat** Freshwater streams and lakes
- **Length** 3 ft (1 m)
- **Diet** Crayfish, shrimp, and other invertebrates

Heliobatis may have been a relative of the stingray.

Its tail contained up to three needlelike stingers that may have been able to inject venom. It lived at the bottom of lakes and possibly rivers, where it hunted for crayfish, small fish, and possibly snails. It was named *Heliobatis* ("Sun ray") because of the way its fins fan out around it like rays of sunlight.

